

**JURISGUARD: SECURE LEGAL DOCUMENT EXTRACTION AND
ARCHIVING SYSTEM**



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In Partial Fulfillment
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BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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CHAPTER 2

METHODOLOGY

In this chapter presents the methodology used in the development of JurisGuard: Secure Legal Document Extraction and Archiving System. It explains how the study is planned, organized, developed, and improved to achieve the objectives of the project. This chapter discusses the hybrid system development life cycle adopted for the study and the major activities involved in the Planning, Analysis, Design, Prototype Testing, Refinement, Implementation, and Maintenance phases. It also describes how the proposed system is prepared for development, testing, evaluation, and future enhancement as a secure platform for legal case information management.

The study adopts a hybrid system development life cycle that combines the structured flow of Waterfall with the iterative and flexible practices of Agile, as shown in Figure 1. This arrangement is appropriate for JurisGuard because the project requires both careful upfront definition of requirements and repeated refinement of selected system features before full implementation. In software engineering literature, plan-driven development is commonly associated with clear sequencing, documentation, and controlled progression across phases, while agile and iterative approaches are valued for feedback, adaptation, and continuous improvement during development [17], [18], [20]. Studies on hybrid development further show that, in practice, many teams use a traditional process model as the overall framework and then integrate fine-grained agile practices where flexibility is needed, which closely matches the logic of the present study [19], [23]. Because JurisGuard must combine OCR-assisted extraction, verification-based encoding, centralized archival storage, and security controls such as RBAC, MFA, encryption, and audit logging, this hybrid structure provides both order and adaptability for the development process.

The project begins with the Planning phase, where the researchers define the scope, objectives, schedule, and resources needed for the development of JurisGuard. In the Analysis phase, the researchers examine the existing handling of legal records, identify current problems and user needs, and determine the functional and non-functional requirements of the proposed system. These requirements include the need for OCR-assisted extraction, a verification interface, centralized case archiving, and security mechanisms that support confidentiality, accountability, and controlled access. In the Design phase, the researchers transform these requirements into a clear structure by preparing the system architecture, process flow, data organization, interface layout, and overall design of the proposed platform. This sequence follows the logic of plan-driven development, where important requirements and designs are clarified before full-scale building begins [17], [19].

After the design phase, the study applies Agile principles through Prototype Testing and Refinement, which are shown in the figure as repeated and connected activities. Iterative and incremental development literature explains that repeated cycles allow teams to learn from earlier versions and improve later versions more effectively, while agile research emphasizes feedback, collaboration, and responsiveness to change [18], [20], [21]. Recent prototyping research also describes prototyping as an established practice in product design, user interface design, and requirements engineering within agile development, making it suitable for projects that need early review and revision of important features [24]. In this stage of the present study, selected parts of JurisGuard, such as OCR-assisted extraction, the verification interface, and access-related functions, are tested in smaller cycles so that errors, usability concerns, and workflow issues can be identified and corrected before full implementation. This approach strengthens the quality of the proposed system by improving accuracy, usability, and fit with the security and workflow needs of legal case information management.

In the Implementation phase, the researchers develop and integrate the major components of JurisGuard based on the approved design and the refined prototype results. This phase includes the actual development of system functions for document extraction, verification-based encoding, case storage, searching, tracking, and security enforcement, followed by testing to confirm that the modules work properly as one system. The final phase is Maintenance, where the researchers evaluate the developed system, identify technical or usability issues, and apply the needed corrections and improvements within the limits of the study period. Although maintenance in a capstone project is naturally limited in duration, it remains important because it helps ensure that the proposed system is stable, functional, and ready for future enhancement. In this way, the hybrid life cycle supports a development process that is both systematic and adaptive, which is appropriate for a secure legal document extraction and archiving system such as JurisGuard [22], [23].

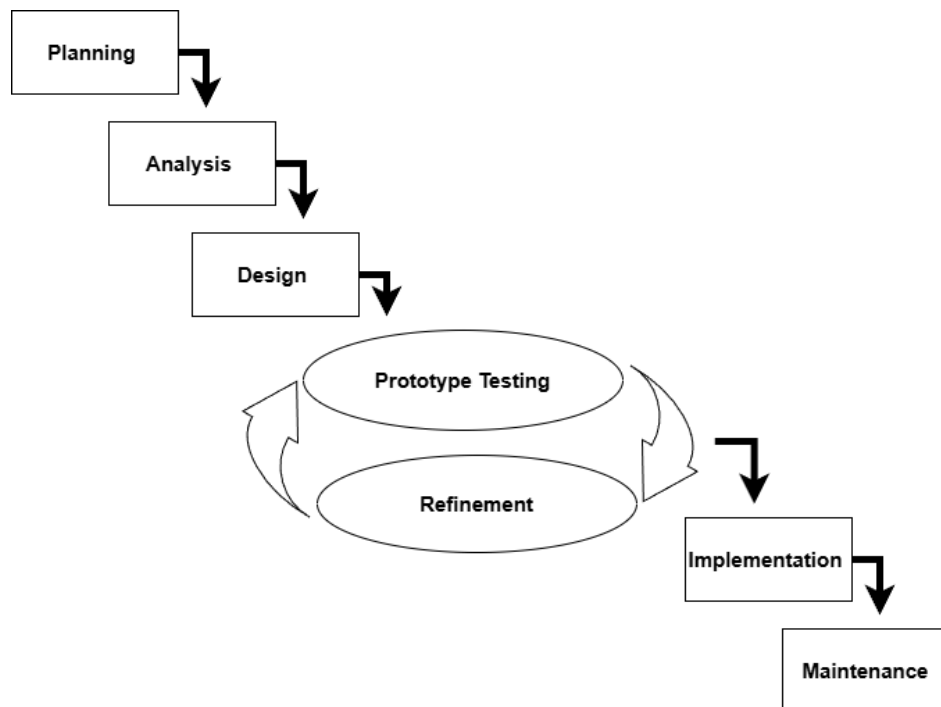


Figure 1. Proposed Hybrid Waterfall-Agile SDLC Model

SYSTEM PLANNING

Project Team Organization

The project team, as shown in Figure 2, is structured to promote clear coordination and proper distribution of responsibilities throughout the development of JurisGuard. The Adviser provides academic and technical guidance to ensure that the study and the proposed system meet the required standards of the capstone project. The Project Manager, together with the Documentation role, is responsible for overseeing the project timeline, coordinating tasks, monitoring progress, and preparing the necessary project documents and reports. The System Analyst focuses on examining user needs, identifying system requirements, and helping define the overall structure and functions of the proposed legal document extraction and archiving system, while the System Developer is responsible for coding, integration, and implementation of the system features. This team structure supports effective collaboration by balancing project management, documentation, system analysis, and technical development in an organized and practical manner.

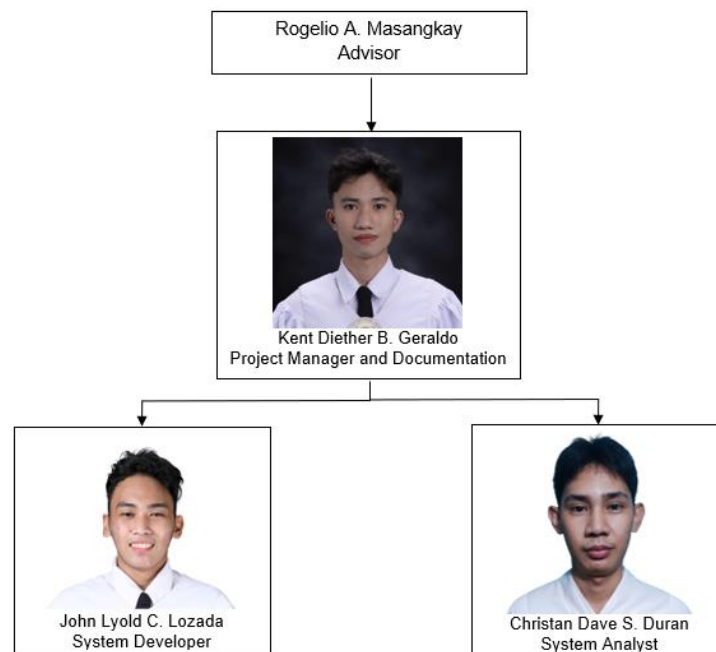


Figure 2. Project Management Team Organization

Work Breakdown Structure

The Work Breakdown Structure (WBS), as shown in Figure 3, presents the development of JurisGuard through seven major phases: Planning, Analysis, Design, Prototype Testing, Refinement, Implementation, and Maintenance. Each phase is divided into specific tasks to ensure that all important activities, from defining objectives and gathering requirements to testing, development, and system improvement, are properly organized. This structure helps the researchers manage the project in a clear and systematic way while ensuring that key features such as OCR-assisted extraction, verification-based encoding, centralized archiving, and security controls are addressed at the appropriate stages. It also supports better task assignment, time management, and coordination among the project team throughout the development process. The WBS serves as a practical guide for completing the proposed system in an organized, efficient, and academically appropriate manner.

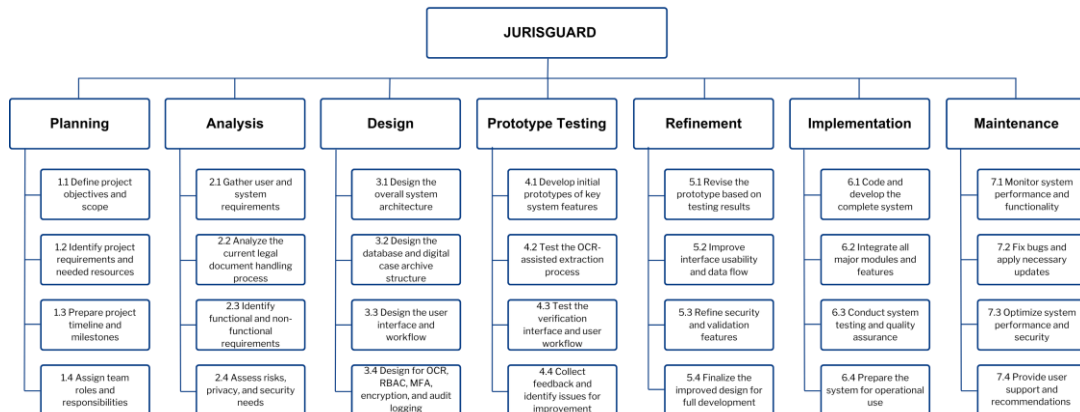


Figure 3. Work Breakdown Structure of JurisGuard Development

Gantt Chart

The Gantt chart in Figure 4 presents the timeline of activities for the development of JurisGuard, showing how each phase of the project is arranged from Planning, Analysis, and Design to Prototype Testing, Refinement, Implementation, and Maintenance. It illustrates the proper sequence and duration of the major tasks involved in developing the proposed legal document extraction and secure archiving system, ensuring that important activities such as requirements gathering, system design, feature testing, refinement, coding, integration, and system evaluation are completed within the project period. By organizing the work into a clear schedule, the chart helps the researchers monitor progress, manage time effectively, and maintain coordination throughout the study. It also serves as a guide for making sure that each phase is completed in the correct order before moving to the next stage of development. The Gantt chart supports the structured, timely, and organized completion of the capstone project.

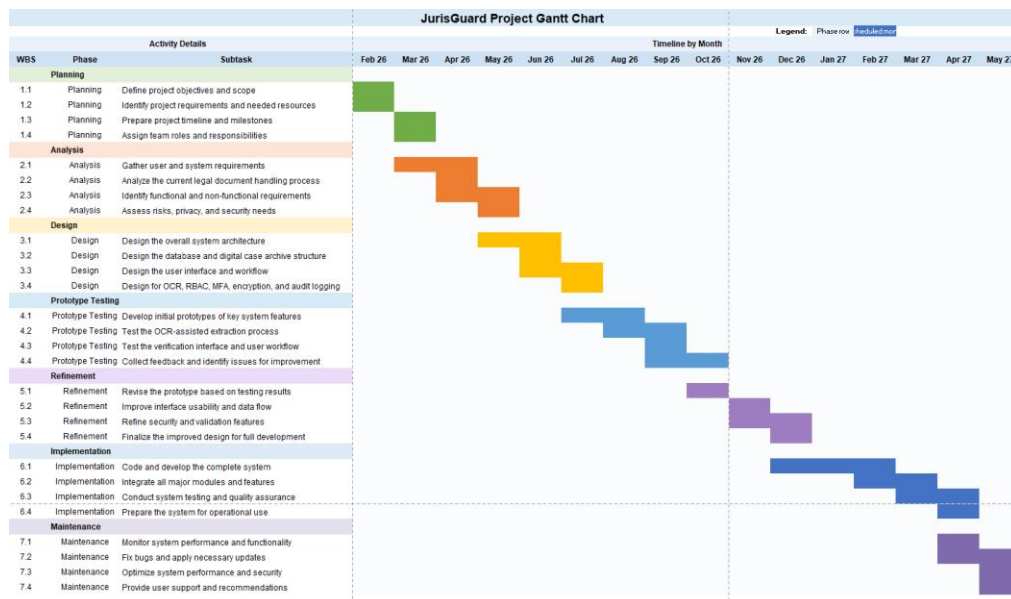


Figure 4. JurisGuard Project Gantt Chart

SYSTEM ANALYSIS

System Architecture

The system architecture of JurisGuard, as shown in Figure 5, presents the main components and overall flow of the proposed legal document extraction and archiving system. It is designed to support secure access, organized case handling, and proper management of legal records through a clear and connected structure. The architecture begins with two authorized users, namely the Admin and Legal Staff, who enter the system through the authentication and role validation process. This part of the system ensures that access is granted based on user role before the user is redirected to the main dashboard, which serves as the central point for accessing the system's major functions.

From the dashboard, the users can access the core modules of the system, which include Client Management, Case Management, and Search and Record Retrieval. The Client Management and Case Management modules support the handling of client information and case records, while the Search and Record Retrieval module allows users to locate stored information quickly and efficiently. These modules are connected to the main record storage of the system, which is the Centralized Case Archive. Through this structure, the system provides an organized way of managing legal records from entry to retrieval while maintaining a clear flow of information across its major functions.

A major part of the architecture is the data entry and document processing flow. When adding client or case information, users may enter data manually or provide a scanned or uploaded document, after which the system performs document text extraction and sends the result to a verification and editing stage before saving the final information. This process is important because it allows extracted details to be reviewed and corrected first, helping improve the accuracy and reliability of the stored records. After

verification, the approved information is saved in the Centralized Case Archive, where it becomes available for future retrieval, updating, and case history monitoring.

The architecture also includes supporting components that strengthen accountability and record monitoring within the system. The Case History Tracking and Updates component helps monitor changes made to case information over time, while the Audit and Activity Log records important user actions and system events. These features support traceability and proper monitoring of record-related activities, which are essential in a system that manages sensitive legal information. Overall, the system architecture of JurisGuard shows a logical and secure flow in which authorized users access the system, manage client and case data, verify extracted information, store records in a centralized archive, and maintain accountability through logging and history tracking.

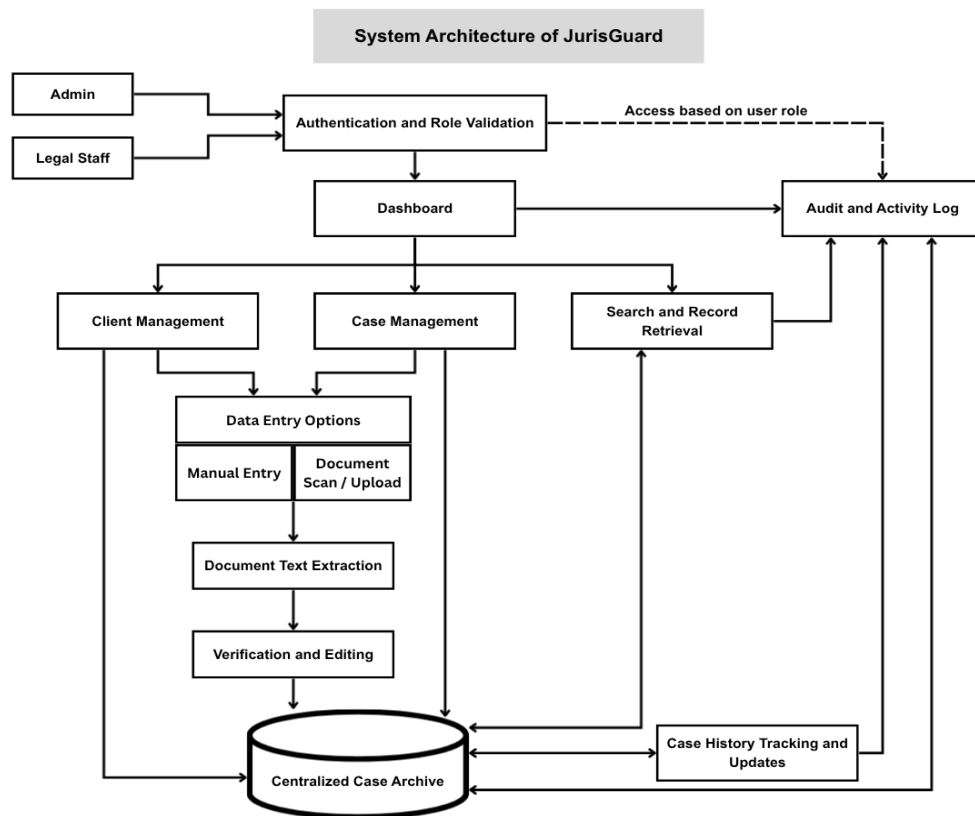


Figure 5. System Architecture of the JurisGuard

Functional and Non-Functional Requirements

The functional and non-functional requirements define the expected capabilities and quality characteristics of JurisGuard as a secure legal document extraction and archiving system. The functional requirements identify the main services the system must perform in support of legal record intake, document processing, storage, retrieval, and monitoring. The non-functional requirements describe the quality conditions that the system must meet to operate effectively, securely, and reliably, following the ISO/IEC 25010 software quality model. Together, these requirements provide a clear basis for system development, evaluation, and future improvement.

Functional Requirements

The system shall provide the following functionalities:

1. The system shall allow only authorized users, specifically the Administrator and Legal Staff, to access the system through a secure login process.
2. The system shall validate user credentials and identify the corresponding user role before granting access to the system.
3. The system shall provide role-based access control so that users can only access functions permitted for their assigned role.
4. The system shall provide a dashboard that serves as the main access point to the system's records and major functions.
5. The system shall allow users to create, view, update, and manage client records.
6. The system shall allow users to create, view, update, and manage case records.
7. The system shall support manual data entry for client and case information.
8. The system shall allow users to scan or upload legal documents for record processing.

9. The system shall extract text and relevant data from scanned or uploaded legal documents.
10. The system shall allow users to review, verify, and edit extracted document data before final submission.
11. The system shall store approved client, case, and document information in a centralized archive.
12. The system shall allow users to search, filter, and retrieve stored records efficiently.
13. The system shall support case history tracking by recording updates and changes made to case records over time.
14. The system shall record important user actions and system events in an audit log for monitoring and accountability.
15. The system shall allow the Administrator to manage user accounts and related access information.
16. The system shall support the handling of printed English and Filipino legal documents within the scope of the study.

The following are the system's non-functional requirements:

1. **Functional Suitability.** The system shall provide functions that are appropriate, complete, and accurate for legal document extraction, verification, case management, archival storage, and record retrieval.
2. **Performance Efficiency.** The system shall process login requests, record submission, document extraction, verification, storage, and retrieval within a reasonable response time that supports daily legal office operations.
3. **Compatibility.** The system shall operate properly within the intended institutional environment and support consistent interaction among its modules for authentication, record handling, document extraction, storage, and monitoring.

4. **Usability.** The system shall provide a clear, understandable, and easy-to-use interface that allows Administrator and Legal Staff users to perform their tasks with minimal confusion and training.
5. **Reliability.** The system shall perform its required functions consistently during login, record processing, storage, retrieval, and audit monitoring, while minimizing errors and data loss during normal use.
6. **Security.** The system shall protect sensitive legal and client information through authentication, role-based access control, secure password handling, controlled record access, and activity logging.
7. **Maintainability.** The system shall be organized in a way that supports correction of errors, modification of features, and future enhancement of system functions without major difficulty.
8. **Portability.** The system shall be capable of being deployed, transferred, or adapted to the intended computing environment of similar legal or institutional offices with minimal changes to its core functions.

Use Case Diagram

As shown in Figure 6, the primary actor in JurisGuard is the **Legal Staff**, while the **Administrator** serves as the secondary actor. The Legal Staff performs the main operational functions of the system, including logging in, accessing the dashboard, managing client records, managing case records, searching and retrieving records, and tracking or updating case history. These interactions show that the Legal Staff is the main user responsible for the day-to-day handling of legal information in the system. On the other hand, the Administrator supports system control and supervision through functions such as user authentication, dashboard access, verification and editing of extracted data, record retrieval, case history monitoring, viewing the audit and activity log, and managing user accounts.

The use case diagram also shows how record handling is performed through the major system processes. When the Legal Staff manages client or case records, the process may involve either manual data entry or scanning or uploading a document, depending on the available source of information. The system then supports document text extraction and allows the extracted information to be verified and edited before it becomes part of the record workflow. In addition, the connection between record retrieval, case history tracking, and audit log viewing shows that the system is designed not only for encoding and storage, but also for monitoring updates and maintaining accountability. Overall, the diagram presents a clear interaction between the Legal Staff, the Administrator, and the main functions of JurisGuard in supporting secure, organized, and traceable legal document management.

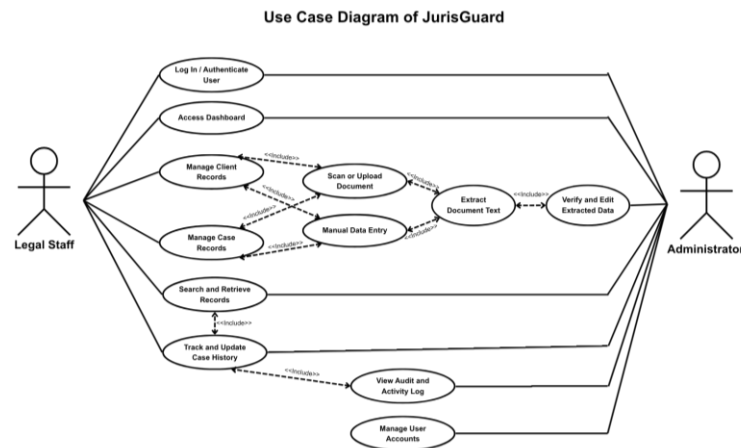


Figure 6. Use Case Diagram of the JurisGuard

Context Flow Diagram

The Context Flow Diagram shown in Figure 7 presents a high-level view of JurisGuard by showing the main external entities connected to the system and the general flow of information between them. At the center of the diagram is JurisGuard, which serves as the secure legal document extraction and archiving system for handling client and case records, supporting document processing, and maintaining organized record storage.

The Legal Staff acts as the primary user by sending credentials, client and case data, manual or scanned inputs, and search or update requests to the system, while the system returns dashboard access, verified records, and retrieved records or case history. The Administrator functions as the secondary user by providing credentials, user management data, and audit requests, and in return receives dashboard access, managed user information, and audit logs for monitoring and control.

The diagram also shows that the document source provides scanned or uploaded documents to the system and receives extraction status as feedback, while the OCR service receives document images from the system and returns extracted text and metadata for further verification. In addition, JurisGuard exchanges data with the centralized case archive database by storing or updating records and retrieving stored data, search results, and other record-related information. These data flows show how information enters, is processed, and is returned by the system to support secure login, record entry, document extraction, verification, retrieval, updating, and audit monitoring. Overall, the Context Flow Diagram helps explain how JurisGuard interacts with its users and supporting components in one organized process without yet showing the detailed internal operations of the system.

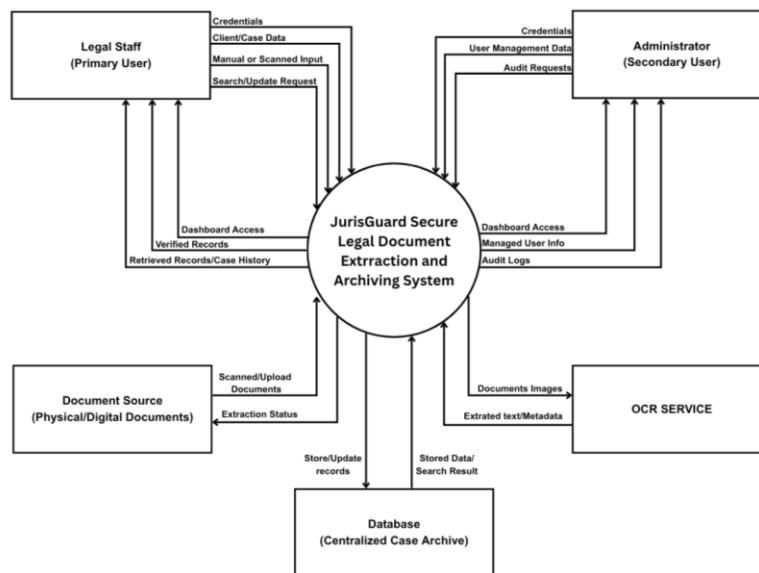


Figure 7. Context Flow Diagram of the JurisGuard

Data Flow Diagram

The Data Flow Diagram shown in Figure 8 presents a more detailed view of JurisGuard by expanding the main system into specific internal processes. While the overall system is identified as 0.0 JurisGuard, the diagram further breaks it down into five major processes: 1.0 Authenticate User and Validate Role, 2.0 Manage Client and Case Data, 3.0 Extract and Verify Document Data, 4.0 Store and Retrieve Records, and 5.0 Monitor Audit and Case History. This structure helps explain how data moves through the system from user access up to record storage, retrieval, and monitoring. As a result, the diagram gives a clearer understanding of how the proposed secure legal document extraction and archiving system works in practice.

The main external entities shown in the diagram are the Administrator, Legal Staff, Document Source, and OCR Service. The Administrator and Legal Staff both provide login credentials to the authentication process and receive access status or dashboard access after validation. The authentication process also checks account information through the D1 User Accounts data store to confirm user identity and role permissions. After successful access, the system allows the users to continue to the next processes for record management and other system functions.

The 2.0 Manage Client and Case Data process handles the input of client and case information coming from the users, including manual entry, search requests, and update requests. It also receives scanned or uploaded documents from the Document Source, then passes the record data to 3.0 Extract and Verify Document Data for document-based processing. In this process, document images are sent to the OCR Service, which returns extracted text and metadata that can be reviewed and verified before the data is accepted. Once the records are approved, they move to 4.0 Store and Retrieve Records, which interacts with D2 Case Records and D3 Case Archive to save, update, and retrieve the needed information.

The diagram also shows that monitoring and accountability are handled through 5.0 Monitor Audit and Case History. This process receives activity data and update data from the storage and retrieval process, then stores or accesses the information through D4 Audit and Case History Log. Through this flow, the Administrator can receive audit logs, monitoring data, and case history or update status, while the Legal Staff can receive retrieved records and case history needed for daily operations. Overall, the diagram explains how JurisGuard manages authentication, record handling, document extraction, storage, retrieval, and monitoring in one connected and organized system.

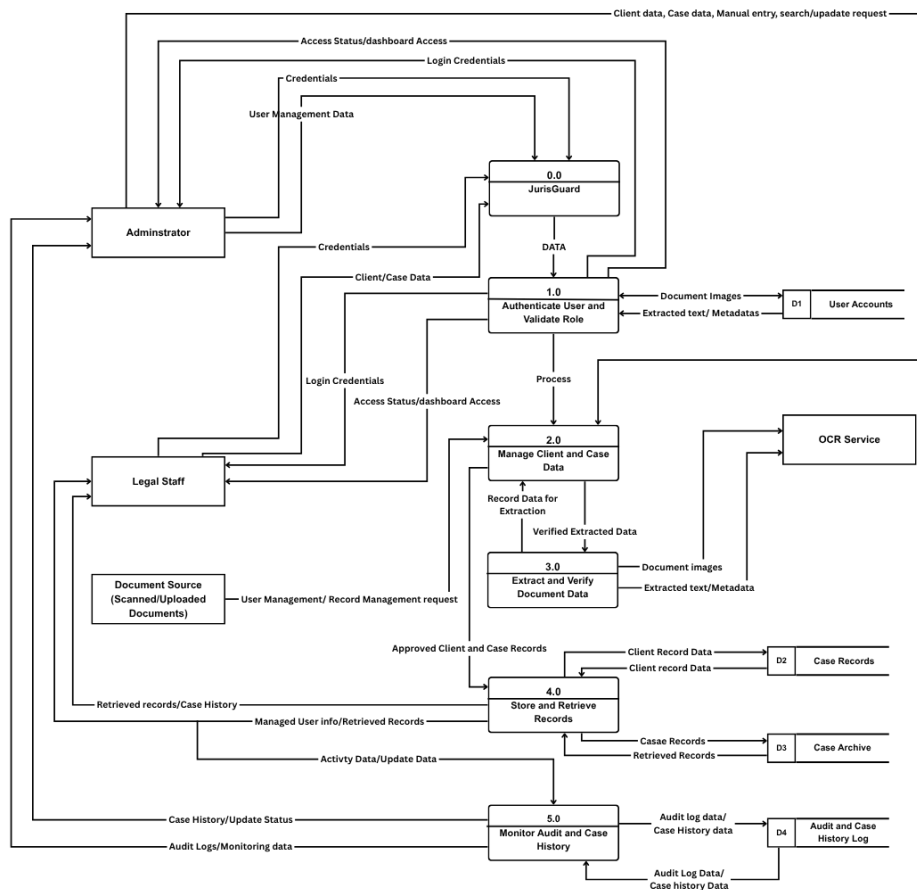


Figure 8. Data Flow Diagram of the JurisGuard

SYSTEM DESIGN

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrated in the JurisGuard system represents the structured database design for the proposed “JURISGUARD: Secure Legal Document Extraction and Archiving System.” It defines how the system’s core entities Users, Roles, Clients, Intake Records, Cases, Documents, and Extracted Metadata are organized and interconnected to support secure legal data management, structured case handling, and intelligent document processing.

The top layer, the user and ROLE entities manage authentication, authorization, and system access control. Each user is assigned a specific role, which determines their permissions within the system. The AUDIT_LOG entity is linked to users to record all system activities such as actions performed, affected records, timestamps, and IP addresses, ensuring transparency, accountability, and traceability of all operations.

The central focus of the system is the CLIENT entity, which stores the personal and demographic information of individuals seeking legal assistance. This is extended through the CLIENT_DETAILS entity, which contains additional contact information, financial data, and detention-related details, while the CLIENT_CLASSIFICATION entity categorizes clients based on legal status, vulnerability, and social classifications. These entities are directly linked to the CLIENT table through one-to-one relationships, forming a complete and structured client profile.

The INTAKE_RECORD entity serves as the main entry point of legal requests and case intake processing. It connects each client to an assigned interviewer (USER) and stores detailed intake information such as nature of request, case facts, proof submissions, and legal declarations. Supporting this process are the REPRESENTATIVE and ADVERSE_PARTY entities,

which capture additional individuals involved in each intake record, establishing one-to-many relationships with the intake process.

For case management, the CASES entity links intake records and clients to legal case data, including case numbers, status, court outcomes, and termination details. It is further categorized through the CASE_NATURE and COURT_BRANCH entities, which standardize legal classifications and court assignments to ensure consistency across all case records.

A key feature of the JurisGuard system is secure document handling and intelligent extraction. The DOCUMENT entity stores uploaded legal files associated with cases and intake records, including encryption paths and OCR processing status. Extracted information is then structured and stored in the EXTRACTED_METADATA entity, which contains JSON-formatted data along with verification status handled by system users. This enables secure archiving, data extraction, and validation of legal documents.

The relationships between entities are defined using primary and foreign keys to maintain data integrity and consistency throughout the system. A single client can have multiple intake records and cases, each intake record may involve multiple representatives and adverse parties, and each case can contain multiple legal documents with extracted metadata. This relational structure ensures that JurisGuard operates as a secure, efficient, and intelligent legal document extraction and archiving system.

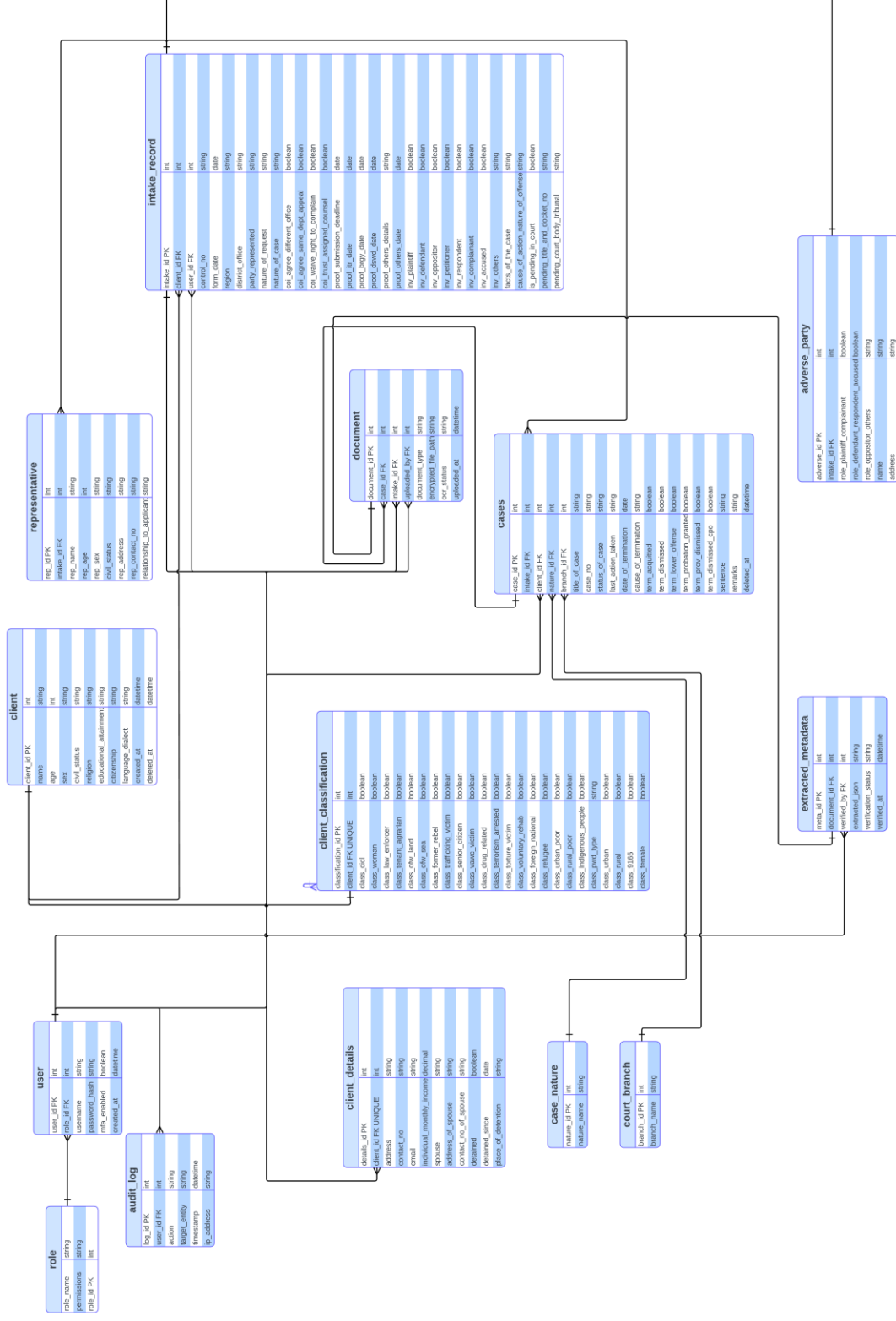


Figure 9. Entity relationship diagram of the JurisGuard

Data Dictionary

The Data Dictionary presents the complete structure of the database used in JurisGuard and serves as a guide for how information is defined, stored, linked, and managed within the system. It identifies the tables, fields, data types, field lengths, constraints, and descriptions that support the main functions of the proposed secure legal document extraction and archiving system. This section is important because it helps make the database more organized, consistent, and understandable during development, testing, and future maintenance. In the case of JurisGuard, the Data Dictionary also supports the study's goal of handling legal case information through centralized storage, OCR-assisted extraction, verification, and secure access control.

The first group of tables supports user access and system control. The ROLE table defines the available roles in the system and includes fields such as role_id, role_name, and permissions, which help determine what actions each type of user is allowed to perform. Related to this, the USER table stores account information such as user_id, role_id, username, password_hash, mfa_enabled, and created_at, making it possible to manage secure login, role assignment, and multi-factor authentication. Together, these tables support the system's requirement for controlled access, role-based permissions, and secure account management for authorized users such as the Administrator and Legal Staff.

The next set of tables focuses on client information, which forms a major part of the system's legal records database. The CLIENT table stores the client's basic profile, including name, age, sex, civil status, religion, educational attainment, citizenship, language or dialect, creation timestamp, and soft delete marker, with some highly sensitive values marked as encrypted. The CLIENT_CLASSIFICATION table extends this by recording legal and social classifications such as CICL, woman, law enforcer, trafficking victim, senior citizen, drug-related classification, indigenous group, disability

type, and other reporting categories. Meanwhile, the CLIENT_DETAILS table stores more specific personal and contact information, including address, contact number, email, monthly income, spouse details, and detention-related information, which further supports the system's need to manage sensitive legal and demographic records in an organized manner.

JurisGuard also includes tables that support intake processing and case management. The INTAKE_RECORD table links the client to the intake transaction and stores fields such as intake_id, client_id, interviewer_id, control_no, form_date, region, party_represented, and nature_of_case. Building on this, the CASE table stores the actual case record using fields such as case_id, intake_id, client_id, nature_id, branch_id, title_of_case, case_no, status_of_case, last_action_taken, date_of_termination, and cause_of_termination. These tables are important because they connect client intake information to the actual legal case being handled, allowing the system to track case status from initial intake up to termination while maintaining a clear relationship between records.

The database structure also supports document management and OCR-assisted processing, which are central features of the study. The DOCUMENT table stores information about uploaded legal documents, including document_id, case_id, intake_id, uploaded_by, document_type, encrypted_file_path, ocr_status, and uploaded_at. After document upload, the EXTRACTED_METADATA table stores OCR-related results such as meta_id, document_id, verified_by, extracted_json, verification_status, and verified_at, making it possible for the system to keep both the extracted data and the verification history. These tables directly support the proposed workflow of scanning or uploading legal documents, extracting text into structured form, and allowing review before final storage.

To support traceability and fuller case context, the database includes several additional tables. The AUDIT_LOG table records important user actions using fields such as log_id, user_id, action, target_entity, timestamp,

and ip_address, which helps the system maintain accountability and activity monitoring. The REPRESENTATIVE table stores information about the client's representative, while the ADVERSE_PARTY table stores information about the opposing or related party in the case, including role indicators and encrypted personal details. These tables strengthen the system's ability to preserve complete legal case information while also supporting secure monitoring of who accessed or changed the records.

Lastly, the COURT_BRANCH and CASE_NATURE tables serve as reference or lookup tables that help standardize important case-related values used in the system. By storing consistent branch and case nature entries, these tables improve data organization and reduce inconsistency during data entry and case management. Overall, the Data Dictionary shows that the database of JurisGuard is not limited to simple record storage, but is carefully structured to support user authentication, client profiling, case intake, document handling, OCR verification, audit logging, and standardized legal record management. In this way, the Data Dictionary helps explain how the database serves as the foundation of the system and supports the overall operation of JurisGuard as a secure and organized legal document extraction and archiving platform.

Field Name	Data Type	Field Length	Constraints	Description
role_id	INT	10	PK, Auto Increment	Unique identifier
role_name	VARCHAR	30	NOT NULL	Name of role
permissions	TEXT		Optional	Role permissions

Table 2: ROLE

Field Name	Data Type	Field Length	Constraints	Description
user_id	INT	10	PK	Unique user ID
role_id	INT	10	FK → ROLE	Role reference

username	VARCHAR	30	UNIQUE, NOT NULL	Login username
password_hash	VARCHAR	30	NOT NULL	Encrypted password
mfa_enabled	BOOLEAN		Default FALSE	MFA status
created_at	DATETIME		NOT NULL	Account creation date

Table 3: USER

Field Name	Data Type	Field Length	Constraints	Description
client_id	INT	—	PK, NOT NULL	Unique identifier of the client
name	VARCHAR	255	NOT NULL	Full name of the client (<i>Encrypted</i>)
age	INT	—	NULL	Age of the client
sex	VARCHAR	10	NULL	Gender (Male/Female)
civil_status	VARCHAR	50	NULL	Civil status
religion	VARCHAR	100	NULL	Religion
educational_achievement	VARCHAR	150	NULL	Highest educational attainment
citizenship	VARCHAR	100	NULL	Citizenship

language_dialect	VARCHAR	100	NULL	Language or dialect
created_at	TIMESTAMP	—	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp
deleted_at	TIMESTAMP	—	NULL	Soft delete timestamp

Table 4: CLIENT

Field Name	Data Type	Field Length	Constraints	Description
classification_id	INT	—	PK, NOT NULL	Unique identifier
client_id	INT	—	FK, NOT NULL	References CLIENT.client_id
class_cicl	BOOLEAN	—	DEFAULT FALSE	Child in Conflict with the Law
class_woman	BOOLEAN	—	DEFAULT FALSE	Woman
class_law_enforcer	BOOLEAN	—	DEFAULT FALSE	Law enforcer
class_tenant_agrarian	BOOLEAN	—	DEFAULT FALSE	Agrarian tenant
class_ofw_land	BOOLEAN	—	DEFAULT FALSE	OFW (land-based)
class_ofw_sea	BOOLEAN	—	DEFAULT FALSE	OFW (sea-based)
class_former_rebel	BOOLEAN	—	DEFAULT FALSE	Former rebel
class_trafficking_victim	BOOLEAN	—	DEFAULT FALSE	Victim of trafficking

class_senior_citizen	BOOLEAN	—	DEFAULT FALSE	Senior citizen
class_vawc_victim	BOOLEAN	—	DEFAULT FALSE	VAWC victim
class_drug_related	BOOLEAN	—	DEFAULT FALSE	Drug-related case
class_terrorism_arrested	BOOLEAN	—	DEFAULT FALSE	Arrested for terrorism-related case
class_torture_victim	BOOLEAN	—	DEFAULT FALSE	Torture victim
class_voluntary_rehab	BOOLEAN	—	DEFAULT FALSE	Under voluntary rehabilitation
class_foreign_national	VARCHAR	100	NULL	Foreign national classification
class_refugee	VARCHAR	100	NULL	Refugee classification
class_urban_poor	VARCHAR	100	NULL	Urban poor classification
class_rural_poor	VARCHAR	100	NULL	Rural poor classification
class_indigenous_people	VARCHAR	100	NULL	Indigenous group classification
class_pwd_type	VARCHAR	100	NULL	Type of disability
class_urban	BOOLEAN	—	DEFAULT FALSE	Urban classification (reporting)
class_rural	BOOLEAN	—	DEFAULT FALSE	Rural classification (reporting)
class_9165	BOOLEAN	—	DEFAULT FALSE	Related to RA 9165
class_female	BOOLEAN	—	DEFAULT FALSE	Female tally classification

TABLE 5: CLIENT_CLASSIFICATION

Field Name	Data Type	Field Length	Constraints	Description
details_id	INT	—	PK, NOT NULL	Unique identifier
client_id	INT	—	FK, NOT NULL	References CLIENT.client_id
address	VARCHAR	255	NULL	Client address (<i>Encrypted</i>)
contact_no	VARCHAR	20	NULL	Contact number (<i>Encrypted</i>)
email	VARCHAR	150	NULL	Email address (<i>Encrypted</i>)
individual_monthly_income	DECIMAL	10,2	NULL	Monthly income (<i>Encrypted</i>)
spouse	VARCHAR	255	NULL	Name of spouse (<i>Encrypted</i>)
address_of_spouse	VARCHAR	255	NULL	Address of spouse (<i>Encrypted</i>)
contact_no_of_spouse	VARCHAR	20	NULL	Contact number of spouse (<i>Encrypted</i>)
detained	BOOLEAN	—	DEFAULT FALSE	Indicates if client is detained
detained_since	TIMESTAMP	—	NULL	Date when client was detained
place_of_detention	VARCHAR	255	NULL	Place of detention

Table 6: CLIENT_DETAILS

Field Name	Data Type	Field Length	Constraints	Description
intake_id	INT	10	PK	Intake record ID
client_id	INT	10	FK → CLIENT	Client reference
interviewer_id	INT	10	FK → USER	User who handled intake
control_no	VARCHAR	20	UNIQUE	Control number
form_date	DATETIME		NOT NULL	Date of intake
region	VARCHAR	50	Optional	Region
party_represented	VARCHAR	50	Optional	Party represented
nature_of_case	VARCHAR	50	Optional	Type of case

Table 7: INTAKE_RECORD

Field Name	Data Type	Field Length	Constraints	Description
case_id	INT	10	PK	Case ID
intake_id	INT	10	FK → INTAKE_RECORD	Linked intake record
client_id	INT	10	FK → CLIENT	Client reference
nature_id	INT	10	FK → CASE_NATURE	Case nature
branch_id	INT	10	FK → COURT_BRANCH	Court branch
title_of_case	VARCHAR	50	NOT NULL	Case title
case_no	VARCHAR	20	Optional	Case number
status_of_case	VARCHAR	20	NOT NULL	Case status
last_action_taken	TEXT		Optional	Case status
date_of_termination	DATETIME		Optional	Termination date

cause_of_termination	TEXT		Optional	Reason for termination
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Table 8: CASE

Field Name	Data Type	Field Length	Constraints	Description
document_id	INT	10	PK	Document ID
case_id	INT	10	FK → CASE	Linked case
intake_id	INT	10	FK → INTAKE_RECORD	Linked intake
uploaded_by	INT	10	FK → USER	User uploader
document_type	VARCHAR		Optional	Type of document
encrypted_file_path	TEXT		NOT NULL	File location
ocr_status	VARCHAR	20	Default pending	OCR status
uploaded_at	DATETIME		NOT NULL	Upload date

Table 9: DOCUMENT

Field Name	Data Type	Field Length	Constraints	Description
meta_id	INT	10	PK	Metadata ID
document_id	INT	10	FK → DOCUMENT	Linked document
verified_by	INT	20	FK → USER	User verifier
extracted_json	JSON		NOT NULL	OCR extracted data
verification_status	VARCHAR	20	Default pending	Status
verified_at	DATETIME	-	Optional	Verification date

Table 10: EXTRACTED_METADATA

Field Name	Data Type	Field Length	Constraints	Description
log_id	INT	10	PK	Log ID
user_id	INT	10	FK → USER	User reference
action	VARCHAR	50	NOT NULL	Action performed

target_entity	VARCHAR	50	Optional	Target table/entity
timestamp	DATETIME		NOT NULL	Action time
ip_address	VARCHAR	20	Optional	User IP

Table 11: AUDIT_LOG

Field Name	Data Type	Field Length	Constraints	Description
rep_id	INT	10	PK	Unique representative ID
intake_id	INT	10	FK → INTAKE_RECORD	Linked intake record
rep_name	VARCHAR	50	NOT NULL	Representative name (encrypted)
rep_age	INT	10	Optional	Age of representative
rep_sex	VARCHAR	20	CHECK (Male/Female)	Gender
civil_status	VARCHAR	50	Optional	Civil status
rep_address	TEXT		Optional	Address (encrypted)
rep_contact_no	VARCHAR	11	Optional	Contact number (encrypted)
relationship_to_applicant	VARCHAR	50	Optional	Relationship to client

Table 12: REPRESENTATIVE

Field Name	Data Type	Field Length	Constraints	Description
adverse_id	INT	10	PK	Unique adverse party ID
intake_id	INT	10	FK → INTAKE_RECORD	Linked intake record

role_plaintiff_complainant	BOOLEAN		Default FALSE	Plaintiff/Complainant role
role_defendant_respondent_accused	BOOLEAN		Default FALSE	Defendant/Respondent/Accused
role_oppositor_others	BOOLEAN		Default FALSE	Oppositor/Other role
name	VARCHAR	50	NOT NULL	Adverse party name (encrypted)
address	TEXT		Optional	Address (encrypted)

Table 13: ADVERSE_PARTY

Field Name	Data Type	Field Length	Constraints	Description
branch_id	INT	10	PK	Unique branch ID
branch_name	VARCHAR	50	NOT NULL	Court or branch name

Table 14: COURT_BRANCH

Field Name	Data Type	Field Length	Constraints	Description
branch_id	INT	10	PK	Unique branch ID
branch_name	VARCHAR	50	NOT NULL	Court or branch name

Table 15: CASE_NATURE

Technologies, Concepts, and Theories

This section presents the key models, algorithms, and technologies used in the development of JurisGuard: Secure Legal Document Extraction and Archiving System. It explains how data is processed from document acquisition up to structured storage and verification, as well as the tools and concepts applied in implementing the system. The discussion follows the major stages of the system workflow, namely data collection, data pre-processing, feature extraction, classification, algorithm selection, and supporting technologies [23], [24], [25], [26].

Data Collection

Data collection in this system begins with the manual ingestion of scanned physical legal records, such as Public Attorney's Office (PAO) interview sheets, affidavits, and court orders. These physical documents are digitized by authorized legal staff and uploaded into the system for further processing. This step ensures that JurisGuard receives the raw image data needed to begin the extraction and archiving workflow while maintaining a controlled and authorized intake process for sensitive legal records [32], [35].

Data Pre-Processing

Before analysis, the collected document images undergo pre-processing to improve image quality and text readability. This stage applies computer vision techniques such as grayscale conversion, binarization, deskewing, and the removal of horizontal lines or other visual noise commonly found in printed legal forms. These operations help produce cleaner input images, which improves the performance of the OCR stage and reduces the likelihood of extraction errors or misread characters [24], [31], [36].

Feature Extraction

Feature extraction in JurisGuard focuses on automated data extraction using Optical Character Recognition (OCR) together with Natural Language Processing (NLP). The pre-processed document images are analyzed to identify relevant text blocks and characters, after which structured patterns such as names, dates, control numbers, and labeled form fields are extracted as metadata features. In addition to OCR, regular expressions are used to capture highly structured values, while NLP techniques help detect meaningful entities from legal text that may not follow a fixed format [23], [25], [28], [36].

Classification

After the text and metadata are extracted, the classification stage organizes the parsed information into structured database fields. For unstructured narrative text, lightweight NLP methods such as Named Entity Recognition (NER) are used to classify terms into useful entities such as persons, locations, and dates. The classified information is then forwarded to a human-in-the-loop verification interface, where legal staff review and validate the extracted values against the original scanned document before the data is committed to the secure archive [23], [25], [28].

Algorithm Selection

EasyOCR was selected as the primary OCR method of the system because OCR performance is critical when dealing with scanned legal forms that may contain degraded text, uneven alignment, or noisy backgrounds. Tesseract remains a lightweight and widely used OCR tool, but prior literature shows that OCR quality can decrease when documents are noisy, handwritten, or not well aligned [31]. PaddleOCR is also a strong alternative because it provides practical multilingual OCR support and efficient document parsing capabilities [26]. However, EasyOCR was chosen for this study because comparative evaluations of open-source OCR engines show that modern deep learning-based OCR approaches perform well on challenging document images, making them more suitable for the type of printed legal records processed in JurisGuard [36].

For the NLP phase, spaCy was selected because it provides efficient and production-oriented Named Entity Recognition for extracting names, dates, and locations from legal narratives [28]. Transformer-based models such as BERT are highly effective for context-rich NLP tasks and complex entity recognition, but they generally require greater computational resources than what is necessary for the targeted metadata extraction tasks of this system [25]. NLTK was also considered because it is a widely recognized NLP toolkit and remains useful for text processing and educational NLP applications, but it was not prioritized for the production-oriented extraction workflow of JurisGuard [23].

Algorithm/Tool	Category	Underlying Architecture	Strengths & Performance	Suitability for JurisGuard
PyTesseract	Optical Character Recognition	Long Short-Term Memory (LSTM)	Fast processing speed and lightweight deployment. Open-source standard but struggles with noisy backgrounds and unaligned text [31].	Moderate. Faster, but lower accuracy compared to deep learning models when reading varied typography on physical legal forms [31].
PaddleOCR	Optical Character Recognition	Differentiable Binarization (DB) & CRNN via PaddlePaddle	Extremely fast and lightweight. State-of-the-art accuracy for multilingual text detection in the wild [26].	High. Excellent alternative, but requires a separate framework ecosystem outside of the project's PyTorch/FastAPI stack [26].

EasyOCR (Selected)	Optical Character Recognition	Convolutional Recurrent Neural Network (CRNN) via PyTorch	High accuracy on degraded documents, complex layouts, and noisy backgrounds. Supports 80+ languages including printed English and Tagalog [36].	High. Ideal for parsing physical PAO forms and IDs where text may be skewed or low-contrast [36].
spaCy (Selected)	Natural Language Processing	Convolutional Neural Networks (CNN) / Pre-trained statistical models	Optimized for production environments. Highly efficient Named Entity Recognition (NER) with fast execution times and low memory footprint [28].	High. Perfect for extracting entities (names, locations, dates) from unstructured legal text like affidavits [28].
BERT	Natural Language Processing	Transformer-based Deep Learning	State-of-the-art accuracy for context-heavy NLP tasks and complex entity recognition [25].	Moderate. Highly accurate but computationally expensive and resource-heavy. Overkill for simple entity extraction in administrative records [25].
NLTK	Natural Language Processing	Rule-based and statistical string processing	Excellent for academic text processing, tokenization, and linguistic research [23].	Low. Too slow and heavily focused on academic research rather than production-level automated extraction pipelines [23].

Table 16: Algorithm Selection

Technologies Used in the System

Computer Vision (OpenCV)

OpenCV is used in the document pre-processing stage of the system. It provides the image processing functions needed to improve scanned legal documents before OCR, including operations such as noise reduction, form-line removal, deskewing, and image enhancement [24].



Figure 10: Computer Vision (OpenCV).

Optical Character Recognition (EasyOCR).

EasyOCR serves as the main OCR engine that converts scanned document images into machine-readable text. Its role in the system is to support the accurate extraction of text from physical legal forms so that the results can be reviewed, verified, and stored in structured form [36].



Figure 11: Optical Character Recognition (EasyOCR).

Natural Language Processing (Regex and spaCy)

Regular expressions are used to capture highly structured values that follow consistent document patterns, while spaCy is used for Named Entity Recognition to identify important metadata such as names, dates, and locations from unstructured legal narratives [23], [28].

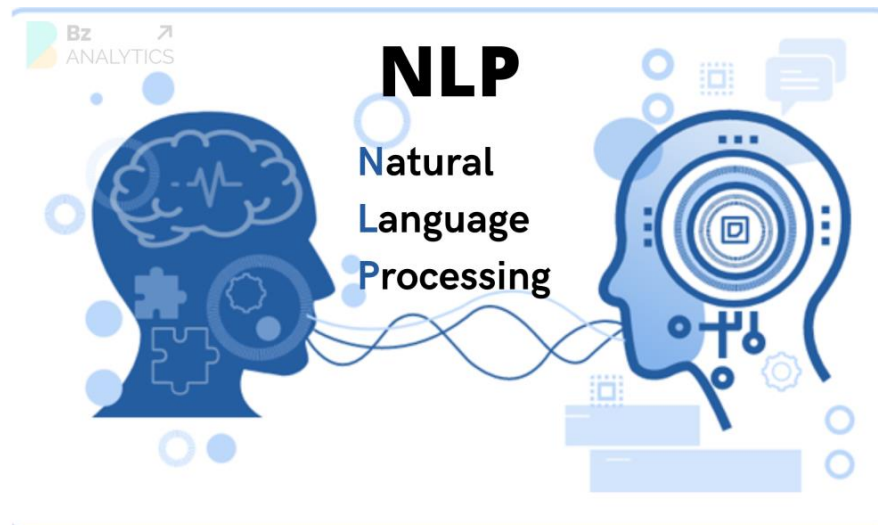


Figure 12: Natural Language Processing (Regex and spaCy)

Backend and API Server (FastAPI).

A Python-based backend framework is used to manage API requests, route extraction tasks, and handle the data exchange between the OCR pipeline, verification process, and database operations. FastAPI is appropriate for this function because it is designed as a high-performance web framework for building APIs with Python [34].

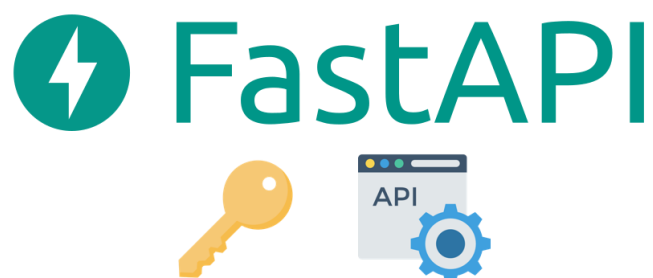


Figure 13: Backend and API Server (FastAPI).

Secure Data Storage (PostgreSQL and pgcrypto).

PostgreSQL serves as the main relational database of the system, while the pgcrypto extension supports cryptographic protection for sensitive stored values. This setup helps JurisGuard protect confidential legal and client information through encryption at rest and secure database handling [33].



Figure 14: Secure Data Storage (PostgreSQL and pgcrypto).

Frontend and Verification Interface (React.js).

React.js is used to build the system's user interface, particularly the verification interface where legal staff review the original scanned document together with the extracted and auto-filled data before final submission. This supports a more interactive and responsive front-end design for record checking, editing, and confirmation. React is appropriate for this role because it is designed for building user interfaces from reusable components and supports web-based client interaction effectively [37].

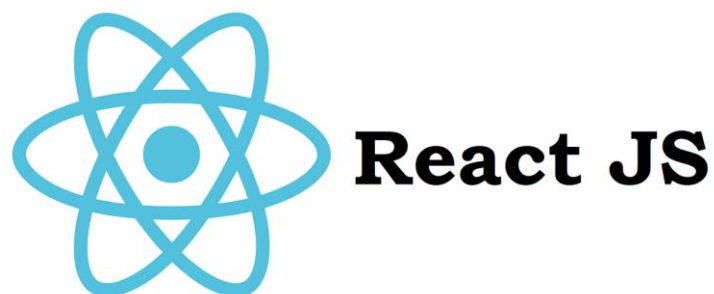


Figure 15: Frontend and Verification Interface (React.js).

SYSTEM TESTING AND IMPLEMENTATION

The system testing and implementation phase focuses on ensuring that JurisGuard operates correctly and meets all specified legal and technical requirements. During this phase, comprehensive testing will be conducted to validate the integration of system components, including the document upload module, the OCR/NLP extraction pipeline [23], [25], [28], [36], data encryption [30], [33], and automated report generation. The testing process will evaluate the performance of the extraction models (EasyOCR, PaddleOCR, and Tesseract) using key metrics such as Character Error Rate (CER), Word Error Rate (WER), and API response time to ensure reliable and timely processing of legal documents [26], [31], [36]. Test cases will be designed to simulate both normal and abnormal conditions, including uploading degraded scans, handwritten inputs, and unauthorized access attempts, to verify that the system responds appropriately and securely [30], [31], [35], [36].

A summary of the testing results will be presented in tabular form, highlighting the performance and status of each system component. To guide the validation process, a System Test Plan will be prepared, outlining the objectives, scope, test environment, test cases, and acceptance criteria.

Following successful testing, the implementation phase will deploy the system within the PAO's Intranet environment. This includes setting up the centralized server, deploying the PostgreSQL database [33], configuring the FastAPI microservices [34], and provisioning the Progressive Web App (PWA) client interface for legal staff [37]. After deployment, users (attorneys, administrative staff, and IT personnel) will be provided with proper training on utilizing the human-in-the-loop Verification Interface to ensure effective system usage.

SYSTEM MAINTENANCE

The System Maintenance phase of JurisGuard will focus on keeping the proposed legal document extraction and archiving system stable, secure,

and functional after implementation. In this phase, the researchers will monitor the system's overall performance, check whether important features such as user access control, document extraction, record storage, retrieval, and audit logging continue to work properly, and address any problems that may affect daily use. Maintenance activities may include correcting system errors, improving data handling, updating user account settings, refining record management functions, and enhancing the accuracy and usability of the verification and retrieval processes. User feedback and evaluation results may also be considered in identifying areas that need improvement, especially in relation to workflow efficiency, ease of use, and the handling of sensitive legal records. Through these activities, the maintenance phase helps ensure that JurisGuard remains reliable, effective, and prepared for future enhancement as a secure case information management system.

System Security Plan

The System Security Plan of JurisGuard defines the safeguards that will be used to protect sensitive client and case records, uploaded legal documents, extracted metadata, user accounts, and audit trails throughout the system's operation. This plan is necessary because the proposed system handles confidential legal information, uses OCR-assisted document intake, stores records in a centralized archive, and relies on role-based access, multi-factor authentication, encryption, and audit logging to support accountability and secure case management. In JurisGuard, the main security priorities are confidentiality, integrity, availability, controlled access, traceability, and the safe handling of record updates from intake up to case termination. These priorities are consistent with the study's focus on secure legal document extraction and archiving for Administrator and Legal Staff users.

The selected basis for this section is ISO/IEC 27001:2022, which is the best-known international standard for information security management

systems and provides a risk-based structure for establishing, implementing, maintaining, and continually improving information security. Its control framework is commonly organized through Annex A domains covering organizational, people, physical, and technological controls, which makes it suitable for a system like JurisGuard that requires both governance measures and technical safeguards for sensitive legal records. For this project, the standard is applied in a practical capstone context by translating relevant control areas into protections for login security, role management, OCR-assisted workflows, encrypted record storage, audit monitoring, backup, and secure handling of legal case data.

The table below aligns JurisGuard with the selected ISO standard by identifying appropriate security domains, control areas, and their practical implementation in the proposed system. The listed measures are based on the actual functions and stored data already defined for JurisGuard, including user accounts, encrypted personal fields, document uploads, extracted metadata, audit logs, and case records.

Security Domain	Control Area	Description / Implementation in the System
A.5 Organizational Controls	Information Security Policy	The system shall follow written security rules for handling client records, case records, uploaded documents, OCR outputs, and audit logs. These rules define access, encoding, verification, and monitoring permissions.
A.5 Organizational Controls	Roles and Responsibilities	Defines clear responsibilities: Administrators oversee accounts and permissions; Legal Staff handle record entry, verification, searching, and updating.
A.5 Organizational Controls	Access Control Governance	Enforces Role-Based Access Control (RBAC) and least-privilege principles. Access rights are reviewed during account creation, modification, or deactivation.
A.5 Organizational Controls	Information Classification	Client identities, legal documents, and metadata are treated as sensitive. Records

		are handled only by authorized users and protected from unnecessary copying.
A.5 Organizational Controls	Information Transfer	Protects data during transmission (login, OCR processing, document upload). Only authorized transfers of legal data are permitted.
A.5 Organizational Controls	Incident Management	Supports procedures for reporting and responding to suspicious logins, unauthorized access, data alteration, or failed retrievals.
A.6 People Controls	Security Awareness & Use	Users are instructed against sharing credentials or bypassing restrictions. Verification of OCR results is required before final submission to ensure accuracy.
A.7 Physical Controls	Workstation Protection	Devices used for JurisGuard must be in controlled areas. Sessions must be locked or terminated when workstations are unattended.
A.7 Physical Controls	Storage & Safeguards	Centralized archive storage is protected from unauthorized physical access or damage. Equipment access is limited to authorized personnel.
A.8 Technological Controls	Secure Authentication	Requires valid credentials and supports Multi-Factor Authentication (MFA). Includes session controls and protection against brute-force attempts.
A.8 Technological Controls	Cryptography	Protects sensitive data via encryption at rest and secure password hashing. Protected file paths and encrypted fields reduce disclosure risks.
A.8 Technological Controls	Logging and Monitoring	Maintains audit logs for logins, record creation, edits, and verifications to ensure accountability and detect suspicious activity.
A.8 Technological Controls	Backup and Recovery	Supports regular backups of all records, metadata, and logs. Recovery procedures are in place for system failures or accidental deletions.
A.8 Technological Controls	Secure Development	System changes and fixes are applied via change control. Input validation and testing are used to mitigate vulnerabilities in OCR and update functions.

A.8 Technological Controls	Vulnerability & Malware	Regular environment checks for security weaknesses. Employs protective measures against malware and unauthorized system modifications.
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Table 17. *System Security Plan*

System Maintenance Plan

The System Maintenance Plan of JurisGuard defines how the proposed system will be supported after development so that it remains stable, secure, and useful for legal record management. This plan is important because JurisGuard handles sensitive client and case information, supports document upload and text extraction, stores records in a centralized archive, and relies on controlled access and audit monitoring for accountability. In this study, maintenance is not limited to fixing errors only, but also includes regular checking of system performance, record handling, security protection, backup activities, and improvements based on actual use and evaluation results. These activities help ensure that the system continues to support secure legal document extraction, verification, storage, retrieval, and monitoring in an organized and reliable manner.

The selected basis for the maintenance plan is ISO/IEC 25010:2023, which is an international standard for systems and software product quality models. According to ISO, this standard defines a product quality model for ICT and software products, and the model is composed of **nine quality characteristics** that can be used to specify, measure, and evaluate product quality throughout the product lifecycle. Because JurisGuard is a software-based legal document extraction and archiving system, ISO/IEC 25010:2023 is an appropriate basis for organizing maintenance activities around quality needs such as reliability, performance efficiency, security, maintainability, usability, compatibility, and related attributes.

In JurisGuard, the maintenance plan is applied in a practical way by linking quality attributes to routine support activities. For example, reliability

is supported through monitoring and backup, security is supported through checking access controls, logs, and protection measures, and maintainability is supported through bug fixing, correction of record-handling issues, and controlled updates to system functions. In the same way, performance efficiency, usability, and compatibility are maintained by reviewing how the system responds during record processing, retrieval, document verification, and user interaction. This approach helps keep the system ready for continued use and future enhancement while staying aligned with the actual purpose and operation of the project.

ISO Attribute	Maintenance Activity	Description	Frequency
Reliability	System Monitoring	Check that login, record entry, document processing, and audit monitoring are functioning without interruption.	Daily
Reliability	Backup and Recovery	Review backup copies of client/case records, metadata, and logs to ensure recovery is possible.	Daily
Performance Efficiency	Performance Monitoring	Review response times and processing speeds for uploads, extractions, and retrievals to identify delays.	Weekly
Security	Security Review & Patching	Check for access-control issues, suspicious activity, and apply security updates to protect sensitive records.	Monthly
Maintainability	Bug Fixing & Debugging	Correct system errors, failed transactions, and incorrect data handling to maintain proper operation.	As needed
Maintainability	Feature Record Updates &	Apply minor improvements to record management and document verification based on user needs.	As needed

Usability	Interface Review	Ensure the UI allows Administrators and Legal Staff to perform tasks (entry, searching, monitoring) efficiently.	Periodically
Compatibility	Environment Checking	Confirm the system functions properly within the institutional environment and with the required hardware.	Periodically
Functional Suitability	Function Validation	Validate that core functions (authentication, extraction, logging) still meet project requirements.	After major updates
Portability	Transfer & Setup Review	Review setup procedures to ensure the system can be easily reconfigured or adapted for other environments.	Before major deployment changes

Table 18. *System Maintenance Plan*

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